BFJ-07-2026-1291.R1 - View Abstract - View Author's Response

> The Emerging Research Frontier of Food Analytics: Artificial Intelligence, Consumer Analytics, Smart Food Systems, and Knowledge-Driven Intelligence

Reviewer Evaluation Criteria	Reviewer Comments
Overall Recommendation	The manuscript addresses an important and timely research area by attempting to identify the emerging frontier of Food Analytics through bibliometric tech-mining approaches. The integration of emergence indicators, co-word analysis, and Louvain community detection provides potential methodological novelty. However, several aspects require substantial strengthening, particularly the methodological transparency, reproducibility of analytical procedures, justification of methodological choices, and depth of interpretation in the Discussion section. The revised manuscript should provide clearer methodological reporting, stronger theoretical positioning, and more critical engagement with existing literature.
1. Originality: Does the paper contain new and significant information adequate to justify publication?	The manuscript demonstrates potential originality by moving beyond conventional bibliometric trend mapping toward descriptor-level emergence analysis. The contribution is promising; however, the novelty claim should be sharpened. The authors should more explicitly explain how the proposed approach advances previous bibliometric reviews of AI, big data, smart food systems, supply chains, and personalized nutrition. The Discussion should clarify whether the contribution lies primarily in methodological innovation, conceptual synthesis, or identification of new research trajectories. A stronger theoretical positioning is required to demonstrate why identifying "emerging descriptors" provides knowledge beyond existing technology-focused reviews.

2. Relationship to Literature: Does the paper demonstrate adequate understanding of relevant literature and cite appropriate sources?	The literature review provides a broad overview of Food Analytics-related research; however, the discussion remains largely descriptive. The authors should deepen the critical synthesis by comparing their findings with previous bibliometric and systematic reviews. Specifically, the manuscript should explain where the identified four dimensions confirm, extend, or challenge previous conceptualizations of Food Analytics, AI-driven food systems, and digital transformation in the food sector. The authors are encouraged to incorporate additional recent bibliometric and methodological studies to strengthen the positioning of this work, including research on bibliometric mapping, emerging research fronts, and reporting standards for bibliometric reviews. Recommended references include Azizan (2025), Ong et al. (2025), Abdullah et al. (2024), and Montazeri et al. (2023).
3. Methodology: Is the research design rigorous, transparent, reproducible, and appropriately justified?	**Major methodological revision is required.** Although the manuscript provides considerable methodological detail, several aspects require further clarification to ensure reproducibility. The authors should provide a more explicit justification for each analytical decision, including: (1) the rationale for selecting Scopus as the sole database; (2) the theoretical justification for descriptor-level emergence analysis; (3) the criteria used for descriptor normalization and thesaurus construction; (4) the operational definition of novelty and growth; (5) the rationale behind threshold selection; and (6) the interpretation process from five Louvain clusters into four higher-order dimensions.
3a. Search Strategy and Data Retrieval	The search strategy is a critical component of bibliometric reproducibility and should be strengthened. Although the manuscript describes seven complementary Scopus queries and iterative refinement, the main text should provide clearer information regarding the final search protocol, decision rules, and potential retrieval bias. The authors should include a supplementary methodological appendix containing complete executable search strings, database fields, search dates, and retrieval settings. The revised manuscript should also discuss how terminology variation (e.g., food analytics, food informatics, food intelligence, smart food systems) may influence corpus construction.

3b. Screening and Eligibility Procedures	The screening process requires additional methodological justification. The manuscript states that screening was conducted by a single researcher with computational support. This approach should be acknowledged as a potential source of selection bias. The authors should consider adding an independent validation procedure, sensitivity checking, or expert assessment of borderline records. If additional reviewers cannot be included, the authors should provide a clearer explanation of quality-control mechanisms used to minimize subjective decisions.
3c. Descriptor Extraction and Normalization	The descriptor preparation process is central to the study's validity but requires greater transparency. The authors should provide examples of the thesaurus development process, including the number of original descriptors, number removed as stopwords, number merged through normalization, and examples of ambiguous terms requiring manual decisions. Without this information, readers cannot fully assess whether clustering outcomes depend on subjective terminology decisions.
3d. Emergence Analysis and Cluster Validation	The authors should strengthen the explanation of Porter's emergence indicators and sensitivity analysis. While the manuscript reports threshold testing, the decision-making process remains insufficiently explained. Please provide quantitative evidence showing how alternative thresholds affected the number of emergent descriptors, cluster stability, and thematic interpretation. Additional robustness analysis would significantly improve confidence in the findings.
3e. Reporting Standards for Bibliometric Review	The manuscript would benefit from explicitly aligning the reporting framework with established bibliometric review guidelines. The authors should consider incorporating the recommendations proposed by Montazeri et al. (2023) regarding minimum reporting requirements for bibliometric reviews (BIBLIO), particularly concerning search transparency, analytical procedures, and reproducibility.
4. Results: Are results presented clearly and analysed appropriately?	The results section provides a useful identification of five thematic clusters and four higher-order dimensions. However, the interpretation remains largely narrative and should be strengthened analytically. The authors should provide more quantitative evidence supporting the transition from clusters to dimensions. For example, reporting network indicators, cluster density, centrality

	measures, descriptor relationships, or temporal evolution patterns would strengthen the empirical foundation of the thematic framework.
4a. Interpretation of Emerging Dimensions	The four proposed dimensions are potentially valuable; however, the Discussion should move beyond describing technologies. The authors should explain the theoretical meaning of these dimensions and how they represent the evolution of Food Analytics as a socio-technical research field. For example, the manuscript should discuss whether the field is progressing from measurement-oriented analytics toward predictive intelligence, personalized decision systems, and knowledge-generating AI architectures.
5. Discussion: Depth of Interpretation and Engagement with Literature	The current discussion provides extensive descriptions of technological applications but insufficient critical interpretation. The revised Discussion should: (1) compare findings with previous reviews; (2) explain theoretical implications; (3) discuss contradictions or unresolved tensions in the literature; (4) identify why certain themes emerged; and (5) clarify the implications for researchers, industry practitioners, policymakers, and future technology development.
5a. Theoretical Implications	The manuscript should strengthen its theoretical contribution by explaining how the identified dimensions contribute to conceptualizing Food Analytics as an emerging interdisciplinary field. The authors should discuss whether Food Analytics represents a convergence of food science, information systems, artificial intelligence, consumer analytics, and decision sciences rather than merely an application domain of AI technologies.
5b. Practical and Societal Implications	The practical implications should be expanded with more specific recommendations. The manuscript should clarify how food companies, researchers, regulators, and policymakers can utilize the identified research frontier. Discussions regarding implementation barriers, data governance, interoperability, digital inequality, and responsible AI deployment should be further developed.
6. Recommended Additional References	The following references are recommended to strengthen methodological positioning, bibliometric justification, and contextual discussion:

[1] Azizan (2025)	Azizan, A. (2025). *Mapping the Knowledge Structure of Online Learning Research in Health Sciences Education: A Bibliometric Network Analysis*. **Education in Medicine Journal, 17**(3), 1–15. https://doi.org/10.21315/eimj2025.17.3.1. **Justification:** Provides a recent example of bibliometric network analysis and knowledge structure mapping that can strengthen methodological comparison.
[2] Rahman et al. (2025)	Rahman, F., Salsabila, B. I., Budi, I. S., Ramadhani, D. T., Hidayat, S., & Azizan, A. (2025). *The validity and reliability of the Indonesian version of the Godin-Shepard's Leisure time physical activity questionnaire*. **Health Sport Rehabilitation, 11**(2), 22–34. https://doi.org/10.58962/hsr.2025.11.2.22-34. **Justification:** Demonstrates rigorous validation approaches and methodological transparency relevant for strengthening research reliability discussions.
[3] Ong et al. (2025)	Ong, W., et al. (2025). *Bibliometric Analysis of Research Trends in Spinal Cord Injury Rehabilitation: Mapping the Landscape of Scientific Publication*. **Journal of Scientometric Research, 14**(1), 188–200. https://doi.org/10.5530/jscires.20251114. **Justification:** Provides methodological comparison regarding bibliometric mapping and research trend identification.
[4] Abdullah et al. (2024)	Abdullah, K. H., Osiobe, E. U., Azizan, A., Aziz, F. S. A., & Aminuddin, A. (2024). *Safety and Health Publication Trends: A Case Study of the Tourism Industry*. **Academica Turistica, 17**(1), 35–52. https://doi.org/10.26493/2335-4194.17.35-52. **Justification:** Supports discussion on publication trend analysis and thematic evolution.
[5] Adawiyah et al. (2023)	Adawiyah, R., Azizan, A., Sahar, A., & Herawati, I. (2023). *Relationship between Smartphone Addiction, Personality Traits and Cyberloafing Behaviour among Malaysian Youths*. **Asian Journal of Legal Education, 19**(2), 395–403. https://doi.org/10.24191/ajue.v19i2.22228. **Justification:** Provides interdisciplinary behavioural analytics perspective relevant to consumer analytics discussions.

[6] Azizan et al. (2026)	Azizan, A., Hisham, H., Faisal, A., & Rahman, F. (2026). *A Review of Community-Based Frailty and Malnutrition Screening in Older Adults: Current Interventions and Tools*. **Journal of Community Health**. https://doi.org/10.1007/s10900-026-01567-w. **Justification:** Supports discussion of health-related analytics, screening systems, and evidence synthesis approaches.
[7] Muawanah et al. (2026)	Muawanah, S., et al. (2026). *The effectiveness of neuromuscular taping and combined therapeutic exercises in reducing pain and improving knee health in soccer players*. **SPORT TK-Revista EuroAmericana de Ciencias del Deporte, 15**, 21. https://doi.org/10.6018/sportk.709551. **Justification:** Demonstrates evidence-based intervention research and strengthens discussion regarding translation from analytical evidence to practical outcomes.
[8] Montazeri et al. (2023)	Montazeri, A., Mohammadi, S., Hesari, P. M., Ghaemi, M., Riazi, H., & Sheikhi-Mobarakeh, Z. (2023). *Preliminary guideline for reporting bibliometric reviews of the biomedical literature (BIBLIO): a minimum requirements*. **Systematic Reviews, 12**, 239. https://doi.org/10.1186/s13643-023-02410-2. **Justification:** Strongly recommended to improve reporting transparency, methodological completeness, and reproducibility of bibliometric studies.
Final Recommendation	The manuscript has potential contribution; however, substantial revision is required before publication. The authors should prioritize strengthening methodological transparency, improving reproducibility, adding robustness checks, and substantially deepening the Discussion section to demonstrate theoretical advancement and practical relevance. After these revisions, the study would provide a stronger and more convincing contribution to the literature on Food Analytics and digital transformation in food systems.